

# LITERATURE REVIEW

## Alzheimer's Disease Detection Using Deep Learning and Explainable AI Team 9

Ahmed Moatasem—ID:2023000917   Omar Zayed — ID: 202300785

Farida Ali — ID: 202301940   Malak Khaled—ID:202300369

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### 1. Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder that gradually erodes cognitive function, memory, and independent living capacity, placing an increasing burden on healthcare systems worldwide. According to the World Health Organization, dementia of which AD accounts for the majority of cases affects more than 55 million people globally, with nearly 10 million new diagnoses each year. Despite this scale, early-stage detection remains a persistent clinical challenge: conventional diagnostic pathways rely heavily on subjective clinical assessment and manual review of neuroimaging data, both of which are labor-intensive and susceptible to inter-specialist variability.

Brain magnetic resonance imaging (MRI) has emerged as the dominant modality for structural assessment of AD-related atrophy, owing to its non-invasive nature and sensitivity to hippocampal and cortical volume loss. However, the subtlety of early morphological changes and the complexity of distinguishing progressive stages (Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented) render manual interpretation unreliable at scale. This motivates the application of deep learning, which can automatically extract discriminative representations from raw MRI data without hand-crafted feature engineering.

Recent years have witnessed a rapid expansion of deep learning approaches for AD classification, ranging from compact custom convolutional neural networks (CNNs) to large transfer-learning pipelines and hybrid architectures that pair deep feature extractors with classical machine learning classifiers. Despite their strong predictive performance, many of these models are opaque, limiting their clinical utility. This has driven growing interest in explainable artificial intelligence (XAI), which seeks to align model decisions with known neuropathological biomarkers, fostering clinician trust.

The present study uses the publicly available Kaggle Alzheimer's Disease Multiclass Dataset (equal-class and augmented version), which provides 2D axial MRI slices across four dementia severity classes. This dataset is particularly suitable because it addresses the pronounced class imbalance present in most AD neuroimaging collections—a known source of bias in deep learning classifiers—through pre-applied augmentation. Against this context, this literature review surveys twelve model-based peer-reviewed studies, compares their architectures, training strategies, and reported performance, identifies their limitations, and justifies the model choices proposed for implementation in this project.

### 2. Deep Learning Architectures for AD Classification

The literature on automated AD classification spans three broad architectural families: custom end-to-end CNNs trained from scratch, transfer learning pipelines built on large pretrained networks, and hybrid models that combine deep feature extraction with classical classifiers. Each family presents distinct trade-offs in terms of data requirements, computational cost, interpretability, and generalisation.

#### 2.1 Custom End-to-End CNNs

Custom CNNs designed specifically for neuroimaging classification offer tight control over architecture depth, regularisation, and augmentation strategy. AbdulAzeem et al. [2] proposed an eight-layer CNN trained end-to-end on the ADNI structural MRI dataset, achieving 99.8% binary accuracy for AD versus cognitively

normal (CN) classification and 97.5% on a multi-class task. The authors demonstrate that careful tuning of convolutional filter sizes, dropout rates, and batch normalisation can compensate for the reduced parameter count relative to deeper transfer-learning models. However, the absence of any explainability mechanism and its restriction to ADNI data limit direct applicability to the Kaggle four-class scenario targeted by this project.

El-Latif et al. [5] further challenged the assumption that architectural depth correlates with performance, proposing a seven-layer lightweight CNN that achieves 99.22% binary accuracy and 95.93% multi-class accuracy on the Kaggle dataset a directly comparable experimental setting to the present work. The key insight is that simpler architectures, when paired with appropriate data augmentation, can match deeper networks while reducing memory and inference cost, which is a practical advantage for clinical deployment. The limitation is that lightweight designs may underfit on more heterogeneous real-world data where richer feature hierarchies are needed.

Ahmed et al. [3] introduced DAD-Net, a five-block custom CNN that directly addresses class imbalance through ADASYN synthetic oversampling, raising accuracy from 90.0% to 99.22% on the Kaggle dataset. This result is significant because it isolates the contribution of the class-balancing strategy: on identical architecture, ADASYN alone accounts for nearly a ten-point accuracy gain. By contrast, the current project's chosen Kaggle variant is already class-equalised through augmentation, reducing but not eliminating this concern. DAD-Net's strength lies in its demonstrated sensitivity to data distribution; its weakness is the absence of any XAI component, which the present project aims to remedy.

Comparing these three custom designs, AbdulAzeem et al. [2] provides the strongest binary result but lacks multi-class generalisation and explainability; El-Latif et al. [5] offers the most practically deployable design for four-class Kaggle classification; and Ahmed et al. [3] contributes the clearest evidence that data balancing strategy is at least as important as architecture choice a lesson directly applicable to this project's preprocessing pipeline.

## 2.2 Transfer Learning and Pretrained Architectures

Transfer learning initializes models with weights pretrained on large image corpora (typically ImageNet) before fine-tuning on medical data, partially mitigating the small-dataset problem endemic to neuroimaging. AlSaeed and Omar [4] demonstrated that ResNet-50's deep residual feature maps, when combined with a Softmax classifier, achieve 99% accuracy on ADNI MRI data, outperforming the same backbone paired with SVM or Random Forest classifiers. This finding that the choice of classification head matters as much as the backbone informs the hybrid experiments described in Section 2.3.

Sava [10] conducted the most systematic transfer-learning comparison in this corpus, evaluating twenty-nine pretrained architectures on the Kaggle AD dataset. EfficientNetB0 emerged as the top performer at 92.98% test accuracy, with EfficientNetB2 and EfficientNetB3 offering superior precision and sensitivity. The EfficientNet family's compound scaling balancing depth, width, and resolution simultaneously proves advantageous over uniform depth-scaling strategies such as VGG or plain ResNets. Nevertheless, the study does not integrate any explainability technique, and the accuracy gap relative to custom CNNs (92.98% vs. 99.22%) suggests that ImageNet pretraining is not universally superior for medical imaging, particularly when domain-specific augmentation is available.

Naz et al. [12] evaluated several standard DNN architectures on the Kaggle MRI four-class task, finding that DenseNet-121 achieved the best average performance across accuracy, precision, recall, and F1-score.

DenseNet's dense connectivity where each layer receives feature maps from all preceding layers promotes feature reuse and gradient flow, which is especially beneficial when training data is limited. The limitation is that dense connectivity increases memory usage substantially relative to ResNet or EfficientNet variants. Mohammed et al. [9] took a hybrid approach, pairing AlexNet feature extraction with an SVM classifier, achieving 94.8% accuracy and a remarkably high AUC of 99.70% on the Kaggle dataset. While AlexNet is architecturally dated, the strong AUC result suggests that SVM boundary optimisation can effectively

compensate for weaker feature representations. This comes at the cost of a two-stage pipeline that is harder to train end-to-end and less amenable to gradient-based XAI techniques.

Basaia et al. [8] demonstrated that a CNN operating directly on single 3D structural MRI volumes, without any hippocampal segmentation or regional preprocessing, achieves 99% AD vs. HC accuracy on ADNI. The clinical significance of this result is its simplicity: eliminating preprocessing steps reduces failure modes in real-world deployment. However, the binary classification scope and 3D volumetric input format are not directly transferable to the 2D axial slice Kaggle dataset used in this project.

## 2.3 Hybrid CNN and Classical Machine Learning Pipelines

Hybrid architectures exploit the complementary strengths of deep feature extractors and classical classifiers: CNNs learn rich spatial hierarchies, while SVMs, Random Forests, and ensemble classifiers are often more sample-efficient and produce decision boundaries that are more stable under data scarcity.

AbdelAziz et al. [1] proposed the SECNN-RF framework, which uses a squeeze-and-excitation CNN (SE-CNN) as a feature extractor feeding into a Random Forest classifier, augmented with SHAP-based explanations. This configuration achieves the highest reported accuracy across the reviewed literature at 99.89%, while simultaneously providing clinically interpretable feature attributions aligned with known AD-related brain regions. The squeeze-and-excitation mechanism recalibrates channel-wise feature responses, allowing the network to suppress task-irrelevant activations before passing representations to the Random Forest. Compared to AbdulAzeem et al. [2]'s plain CNN, the SE-CNN backbone extracts more discriminative representations; compared to AlSaeed and Omar [4]'s ResNet-50+Softmax, the Random Forest classifier provides better handling of complex, non-linear decision boundaries without overfitting. The integration of SHAP also sets this paper apart from all other reviewed works as the only model to pair top-tier accuracy with quantified explainability. Its primary limitation is computational cost: training both the SE-CNN and the Random Forest is more expensive than end-to-end methods, and SHAP computation scales poorly with input dimensionality.

Bangyal et al. [10] proposed the CAD-ALZ framework, pairing a ConvMixer backbone which operates entirely on patch embeddings through depthwise separable convolutions with a Random Forest classifier, and achieving an F1-score of 99.61% and sensitivity of 99.69% on the Kaggle dataset. ConvMixer's patch-based design is architecturally distinct from standard convolutional pipelines, processing global patch relationships at every layer rather than progressively building local receptive fields. This makes it highly effective for capturing both fine-grained cortical texture and global atrophy patterns simultaneously. Compared to SECNN-RF [1], CAD-ALZ achieves comparable F1-score but reports no XAI integration; compared to AlexNet+SVM [9], it provides superior sensitivity through a more expressive feature extractor.

Sharma et al. [11] proposed HTLML, a hybrid transfer-learning and machine-learning ensemble that combines feature extraction from multiple pretrained CNN backbones (including InceptionV3 and VGG16) with a permutation-based voting classifier, achieving approximately 95.23% multi-class accuracy on the Kaggle dataset. The ensemble strategy reduces the variance associated with any single backbone and provides implicit model diversity. However, HTLML's accuracy is notably lower than SECNN-RF [1] and CAD-ALZ [7], likely because the voting classifier does not leverage the discriminative power of individual backbones as effectively as a trained Random Forest. The absence of XAI is also a limitation shared with most reviewed hybrid approaches.

## 3. Multimodal Deep Learning Approaches

Multimodal frameworks fuse complementary imaging modalities typically structural MRI and functional FDG-PET to capture both anatomical atrophy and metabolic dysfunction associated with AD. Huang et al. [6] developed a 3D CNN that fuses T1-weighted MRI and FDG-PET data focused on the hippocampal region,

achieving approximately 91% accuracy for AD versus CN classification on ADNI. The fusion strategy concatenates volumetric feature maps from both modalities before the fully connected classification layers, allowing the model to exploit structural and functional signals jointly.

Compared to the single-modality 3D CNN of Basaia et al. [8], which achieves 99% accuracy on the same ADNI dataset, the multimodal approach of Huang et al. [6] achieves a lower overall accuracy despite incorporating richer information. This apparent paradox likely reflects the challenge of aligning MRI and PET volumetric spaces and the added noise introduced by imperfect fusion. Nevertheless, the multimodal approach is more clinically informative and better aligned with established neuropathological knowledge, which values both structural and metabolic biomarkers. For the present project's 2D Kaggle dataset, direct multimodal fusion is not feasible; however, the insight that anatomically targeted feature extraction (hippocampal focus) improves model specificity informs the XAI validation strategy adopted in this study.

## 4. Explainability in Medical AI (XAI)

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The opacity of deep learning models is a fundamental barrier to clinical adoption. Without interpretable explanations of model decisions, clinicians cannot verify whether predictions are grounded in neuropathologically meaningful features or spurious correlations. The reviewed literature reveals a spectrum of explainability integration: from papers that entirely omit XAI [2, 3, 5, 6, 7, 8, 9, 10, 12] to those that treat explainability as a primary design objective [1].

AbdelAziz et al. [1] stands alone among the reviewed papers in integrating SHAP (SHapley Additive exPlanations) directly into the SECNN-RF pipeline, providing feature-level attribution scores that identify which voxel regions most strongly influence classification. SHAP's game-theoretic foundation guarantees consistency and local accuracy properties absent in gradient-based alternatives such as Grad-CAM. The limitation is computational: SHAP values for high-dimensional MRI inputs require substantial time to compute. In contrast, gradient-based methods like Grad-CAM, while faster and easier to visualise, are architecture-sensitive and can produce unstable saliency maps across similar inputs, reducing clinical trustworthiness.

The broader literature confirms that XAI is increasingly expected not optional in medical AI systems. Papers without explainability, regardless of their accuracy, face growing resistance from clinical communities and regulatory bodies. This reinforces the present project's commitment to integrating multiple complementary XAI methods Grad-CAM, SHAP, LIME, and Integrated Gradients rather than relying on a single technique, and to validating that highlighted regions correspond to established AD biomarkers such as the hippocampus, entorhinal cortex, and parietal lobe.

## 5. Dataset: Kaggle Alzheimer's Disease Multiclass Dataset

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This project uses the Kaggle Alzheimer's Disease Multiclass Dataset (equal-class and augmented version), which contains 2D axial brain MRI slices labelled across four classes: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. The dataset's distinguishing feature relative to other Kaggle AD collections is its pre-applied class equalization : augmentation techniques have been used to ensure approximately equal class representation, directly mitigating the severe imbalance particularly in the Moderate Demented class that degraded performance in earlier studies using unbalanced Kaggle data [3,5].

Several reviewed papers including El-Latif et al. [5], Ahmed et al. [3], Mohammed et al. [9], Sava [10], Naz et al. [12], and Sharma et al. [11] have used versions of the Kaggle MRI dataset, establishing a strong baseline for comparison. However, none of these studies used the specific augmented and equalised variant employed here, which means direct accuracy comparison requires caution. The 2D axial slice format is well-suited to standard 2D CNN architectures, making the models reviewed in Sections 2 and 3 directly

applicable without volumetric preprocessing. Class-weighted loss functions will be used as a secondary safeguard against residual imbalance, complementing the dataset's pre-applied augmentation.

## 6. Gaps in the Literature and Contribution of This Study

The reviewed literature reveals four persistent gaps. First, the majority of high-accuracy models including AbdulAzeem et al. [2], El-Latif et al. [5], Ahmed et al. [3], and Basaia et al. [8] report no explainability mechanism whatsoever, making it impossible for clinicians to verify the anatomical grounding of predictions. This is particularly problematic given that some reported accuracies (above 99%) may reflect dataset-specific overfitting rather than genuine generalisation.

Second, where hybrid architectures exist (e.g., CAD-ALZ [7], HTLML [11], AlexNet+SVM [9], ResNet-50+Softmax [4]), they are evaluated in isolation rather than systematically compared against end-to-end alternatives on the same dataset and under the same experimental conditions. This makes it difficult to attribute performance differences to the classifier choice versus the feature extractor versus the training strategy.

Third, the treatment of class imbalance is inconsistent across the literature. DAD-Net [3] demonstrates that ADASYN oversampling alone can close a ten-point accuracy gap, yet most other reviewed papers apply no explicit balancing strategy and report inflated accuracy on majority-class predictions. The present study's use of the equalised Kaggle dataset, combined with class-weighted loss functions, addresses this more rigorously than any single reviewed paper.

Fourth, no reviewed paper comprehensively combines a high-accuracy hybrid pipeline with multiple complementary XAI techniques validated against known AD biomarkers. AbdelAziz et al. [1] comes closest by integrating SHAP into SECNN-RF, but relies on a single XAI method and does not compare its explanations against clinical literature systematically.

In response to these gaps, the present study proposes an integrated framework that: (i) evaluates multiple model families custom CNNs [2, 3, 5], hybrid pipelines [1, 7, 11], transfer-learning backbones [4, 9, 10, 12], and multimodal approaches [6, 8] under identical experimental conditions on the equalized Kaggle dataset; (ii) applies class-weighted loss functions to guard against residual imbalance; (iii) integrates Grad-CAM, SHAP, LIME, and Integrated Gradients to provide multi-perspective model interpretation; and (iv) validates highlighted regions against established AD neuropathology. This combination of breadth, rigor, and interpretability distinguishes the present study from all individually reviewed works.

## 7. Justification of Selected Models

Table 1 summarises the twelve models reviewed. The eight primary models selected for implementation were chosen to maximise architectural diversity while ensuring that each offers a clear, implementable design backed by strong empirical performance. The four additional references provide supplementary comparative context. The selection rationale for each implemented model is as follows.

| Ref  | Model                   | Architecture Type                | Dataset    | Best Accuracy     | XAI Used |
|------|-------------------------|----------------------------------|------------|-------------------|----------|
| [1]  | SECNN-RF                | Hybrid CNN + Random Forest       | Kaggle MRI | 99.89%            | SHAP     |
| [2]  | Custom 8-layer CNN DAD- | End-to-end Custom CNN Custom     | ADNI       | 99.80%            | None     |
| [3]  | Net + ADASYN 7-layer    | CNN + Oversampling Lightweight   | Kaggle MRI | 99.22%            | None     |
| [4]  | Lightweight CNN CAD-ALZ | Custom CNN                       | Kaggle MRI | 99.22%            | None     |
| [5]  | (ConvMixer+RF)          | Hybrid ConvMixer + Random Forest | Kaggle MRI | 99.61% F1         | None     |
| [6]  | 3D CNN (single MRI)     | End-to-end 3D CNN                | ADNI       | 99.00%            | None     |
| [7]  | HTLML Voting Classifier | Hybrid TL + ML Ensemble          | Kaggle MRI | ~95.23%           | None     |
| [8]  | Multi-modal 3D CNN      | Multi-modal 3D CNN (MRI+PET)     | ADNI       | ~91.00%           | None     |
| [9]  | ResNet-50 + Softmax     | Transfer Learning                | ADNI       | 99.00%            | None     |
| [10] | AlexNet + SVM           | Hybrid TL + SVM                  | Kaggle MRI | 94.80%            | None     |
| [11] | EfficientNetB0          | Transfer Learning (29 models)    | Kaggle MRI | 92.98%            | None     |
| [12] | DenseNet-121            | Transfer Learning                | Kaggle MRI | Best avg. metrics | None     |

## 8. Conclusion

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This review has examined twelve peer-reviewed, model-based studies on deep learning for Alzheimer's disease classification, comparing their architectures, training strategies, class-imbalance handling, dataset choices, and explainability integration. The analysis reveals a clear progression from simple custom CNNs through hybrid deep-classical pipelines to multimodal and transfer-learning frameworks, with accuracy consistently above 99% for the best configurations on both ADNI and Kaggle datasets. However, accuracy alone is an insufficient criterion for clinical adoption: the near-universal absence of XAI mechanisms across reviewed papers represents the most critical unaddressed gap in the field.

The proposed framework addresses this gap by combining the strongest hybrid architecture from the reviewed literature (SECNN-RF [1]) with four complementary XAI techniques applied to the equalised Kaggle four-class dataset. By evaluating multiple model families under identical conditions and validating explanations against established AD biomarkers, this study is designed to advance both diagnostic accuracy and clinical interpretability demonstrating that technical performance and transparency are not competing objectives but mutually reinforcing ones.

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